



SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.

AI-Matched University–Industry Capstone Collaboration

Platform with Skill-Gap Recommendation

CSA0925- JAVA PROGRAMMING

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Supervisor

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SIMATS Engineering

REVIEW-1 FEEDBACK

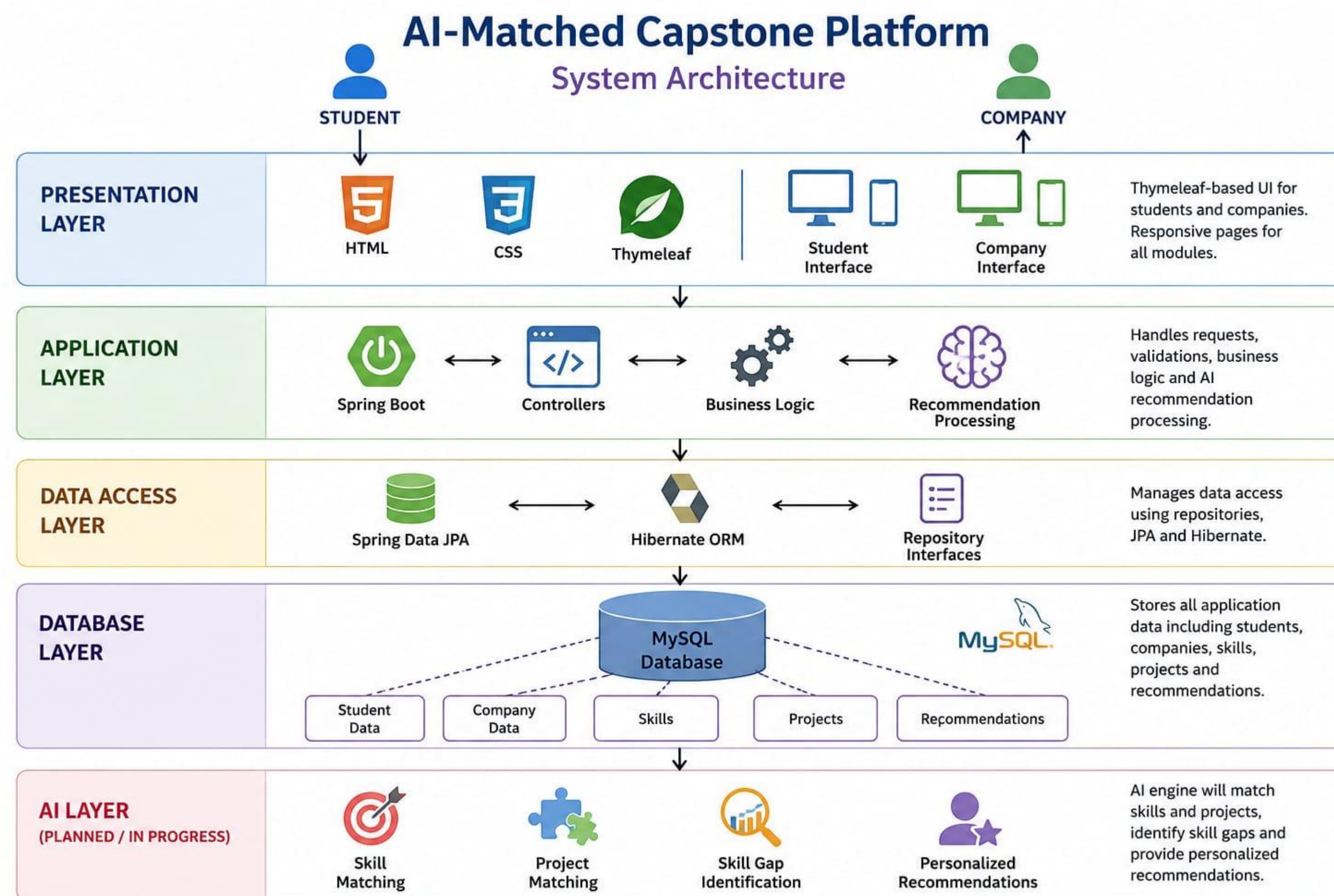
Feedback

- Improve system architecture and module interaction.
- Provide separate Student & Company access.
- Personalize the student dashboard.
- Add AI project recommendations.
- Add Skill Gap Analysis.
- Connect Frontend → Backend → MySQL.
- Improve module navigation.

Corrective Actions

- Implemented Spring Boot + Thymeleaf frontend/backend.
- Connected MySQL using Spring Data JPA/Hibernate.
- Created Student Login, Dashboard, Recommendations, Skills, Projects & Skill Gap pages.
- Added student profile details and personalized dashboard.
- Started implementation of AI Recommendation & Skill Gap modules.

UPDATED ARCHITECTURE



MODULE DESIGN

Module 1 — Student & AI Recommendation Module

- Student Registration & Login
- Student Profile Management
- Skills & Interests
- AI Project Recommendation
- Skill Gap Analysis
- Personalized Dashboard

Purpose:

Matches student skills, interests and academic profile with suitable capstone projects and identifies missing skills.



MODULE DESIGN

Module 2 — Company & Project Management Module

- Company Registration & Login
- Company Profile Management
- Post Capstone Projects
- Define Required Skills
- Manage Project Details
- View Student/Project Matching Information

Purpose:

Allows companies to post projects, specify required skills and manage project-related information.



MODULE DESIGN

Module 3— Collaboration & Progress Tracking Module

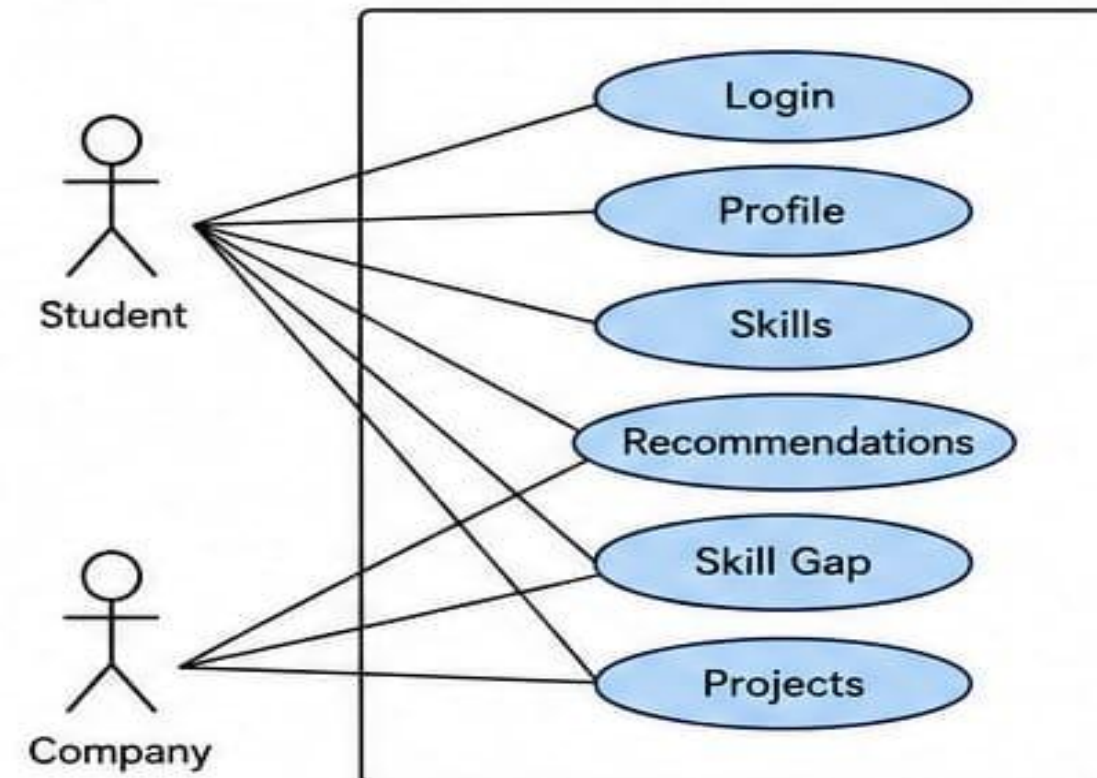
- Facilitates communication among students, faculty, and industry mentors.
- Tracks project milestones, tasks, and deadlines.
- Allows mentors to provide feedback and performance evaluations.
- Monitors student progress through reports and dashboards.
- Sends notifications for meetings, submissions, and updates.
- Generates final project progress and performance reports.

Purpose:

Progress Tracking Module enables seamless communication between students, faculty, and industry mentors while monitoring the progress of capstone projects from start to completion.

UML & DB

Use Case Diagram

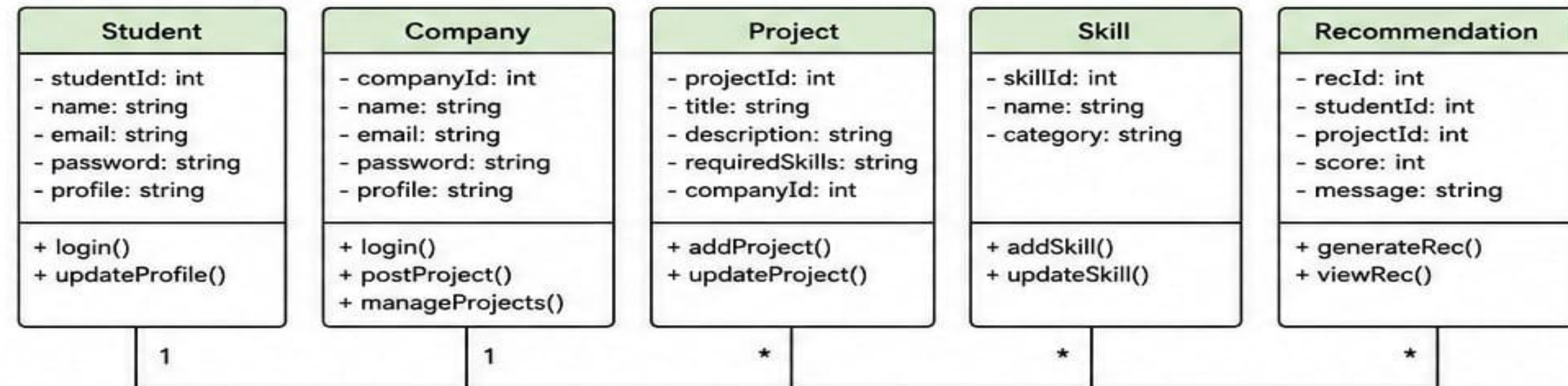


Actors: Student, Company

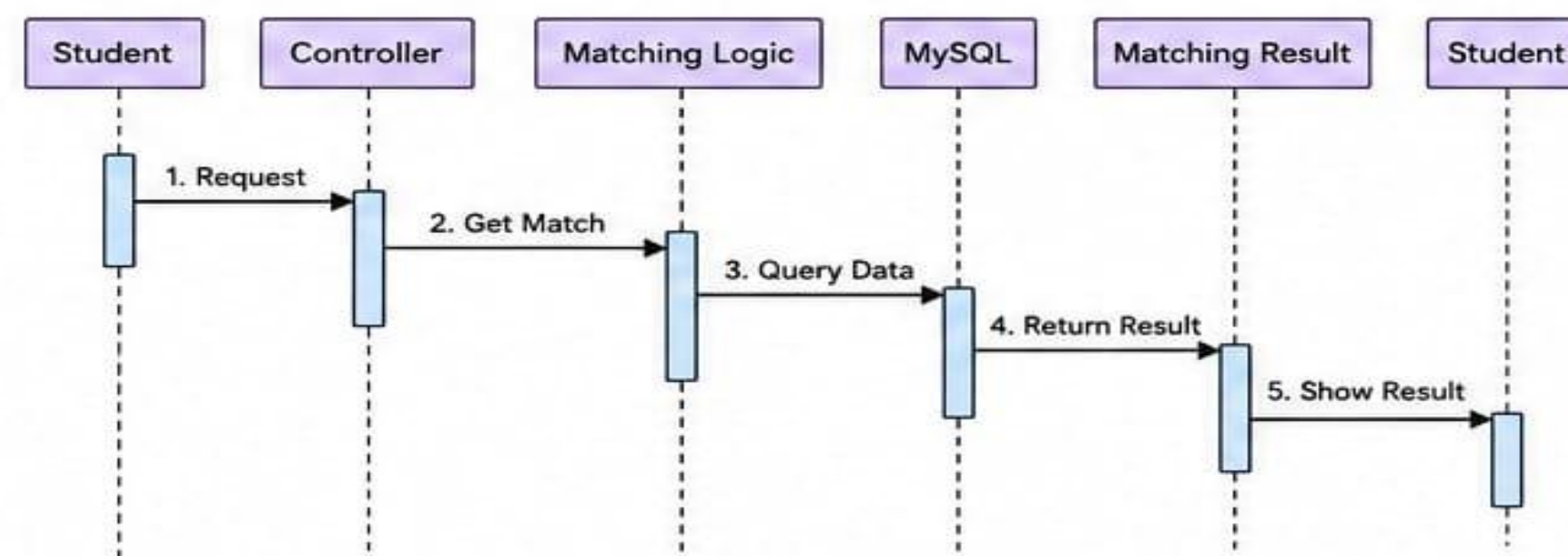
Student: Login, Profile, Skills, Recommendations, Skill Gap, Projects

Company: Login, Profile, Post Projects, Required Skills, Manage Projects

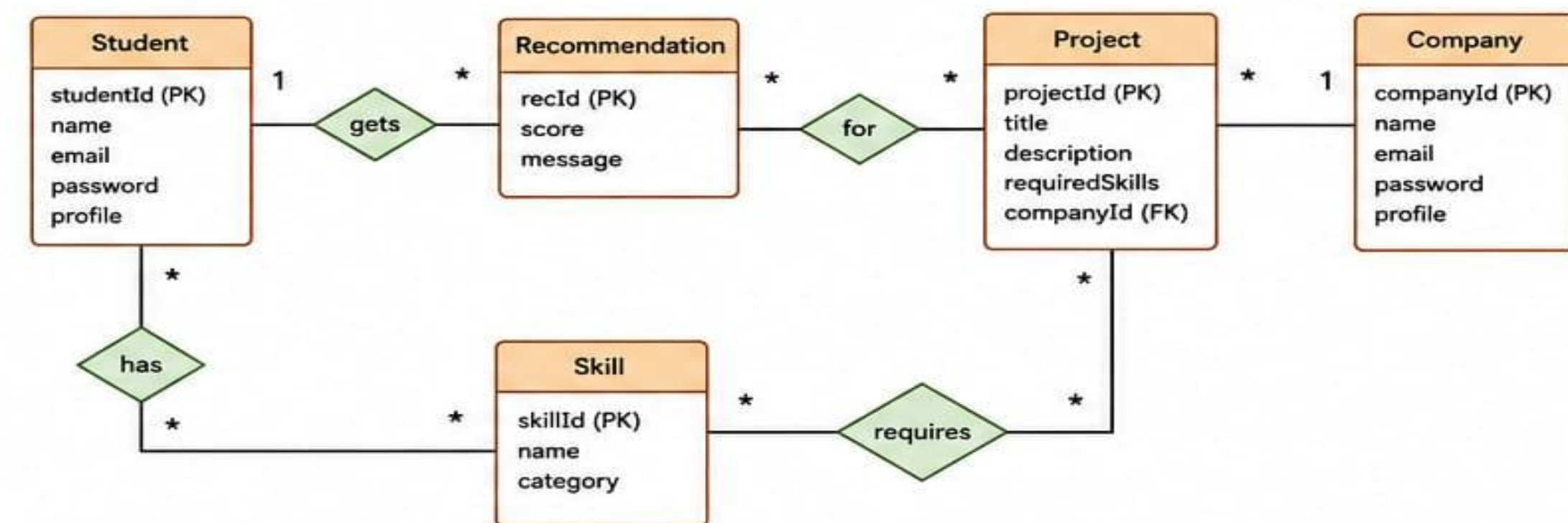
Class Diagram



Sequence Diagram



ER Diagram



TECHNOLOGY STACK

Languages

- **Java 17** — Backend development
- **HTML5** — Web page structure
- **CSS3** — UI styling

Frameworks

- **Spring Boot 4.1.0** — Backend & application development
- **Thymeleaf** — Dynamic web pages
- **Spring Data JPA** — Database interaction

Libraries / Technologies

- **Hibernate ORM** — Object-relational mapping
- **MySQL Connector/J** — Java–MySQL connection
- **Bean Validation** — Input validation

Database

- **MySQL 8.0** — Stores student, company, project and skill data

Development Tools

- **IntelliJ IDEA** — Java/Spring Boot development
- **MySQL** — Database management
- **Maven** — Dependency & project management

Current Implementation

- **Student Login** — Student authentication interface
- **Student Dashboard** — Displays student profile and academic information
- **My Skills** — Displays student's technical skills
- **Projects** — Project listing module
- **AI Recommendations** — Recommendation interface created
- **Skill Gap Analysis** — Skill matching and gap display interface created
- **Backend** — Spring Boot controllers handle student module requests
- **Database** — MySQL connected using Spring Data JPA and Hibernate
- **Frontend–Backend Integration** — Thymeleaf connects the UI with Spring Boot

AI Implementation — In Progress

- Skill-based project matching
- Interest-based recommendation
- Academic-profile-based matching
- Skill-gap identification
- Recommendation score calculation

Next Implementation

- Complete AI recommendation logic
- Connect recommendations with actual database data
- Complete **Company module**
- Improve student–company interaction
- Complete end-to-end testing



IMPLMENTATION



AI Capstone

HomeStudent LoginStudent RegisterCompany LoginCompany Register

AI-Matched Capstone Platform

Find the right capstone project based on your skills, interests, technologies and academic goals using intelligent AI-based matching.

Student

Find capstone projects that match your skills, interests and academic profile.

Student Login

Student Register

Company

Post real-world projects, specify required skills and find students who match your project requirements.

Company Login

Company Register

AI Matching

Get project recommendations based on student skills and interests.

Personalized

Receive capstone suggestions tailored to each student's academic profile.

Company Projects

Companies can create projects and specify the skills required for each project.

Smart Analysis

Compare student skills with project requirements to identify suitable matches.

Student Login

AI-Matched Capstone Platform

Email

Enter your email

Password

Enter your password

Login

Don't have an account? [Create Account](#)

Company Login

Company Email

Enter company email

Password

Enter password

Login

Don't have a company account? [Register Company](#)



IMPLMENTATION



Welcome, jeevana 🙌

Welcome to your personalized AI Student Dashboard.

🎯 AI Recommendations

Get personalized project and opportunity recommendations based on your skills and interests.

View Recommendations

🧩 My Skills

View the technical skills associated with your student profile.

View Skills

🚀 Projects

Explore projects that match your department, interests and skills.

Explore Projects

👤 Student Profile

Name:	jeevana
Email:	student@gmail.com
Department:	CSE
Year:	3
Skills:	java,python
Interests:	AI,ML



IMPLMENTATION



←→↻ ⓘ http://localhost:8080/student/recommendations

AI Recommendations

Personalized project recommendations based on your skills, interests and academic profile.

Welcome, Jeevana!

AI-Based Project Recommendation

The AI recommendation engine analyzes your academic profile, technical skills, interests and project requirements to recommend suitable capstone projects.

AI-Based Student Performance Prediction System

This project is recommended based on your current technical skills, academic background and interests.

★ Project Suitability: 70%

Personalized Suggestions

Based on your current profile, the system recommends improving the following areas:

- Improve Machine Learning fundamentals
- Practice Data Analysis using Python
- Learn Scikit-learn for machine learning projects
- Improve SQL and data handling skills
- Build practical AI/ML projects

Skill Gap Analysis

The system compares your current skills with the skills required for the recommended project. It identifies your matching skills, missing skills and the areas you need to improve.

[View Skill Gap Analysis](#)

Explore Your Profile

Explore your current skills and available capstone projects.

[View My Skills](#) [Explore Projects](#)

AI Capstone

Skill Gap Analysis

Our AI system compares your current skills with the skills required for recommended capstone projects and identifies the skills you need to improve.

Recommended Project

null%

Project Skill Match

Matching Skills

No matching skills identified yet.

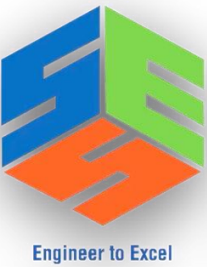
Skill Gaps Identified

No skill gaps identified yet.

Recommended Skills to Learn

[← Back to Recommendations](#) [Explore Projects](#) [Dashboard](#)

AI Matched Capstone Platform
Intelligent Project Recommendation & Skill Gap Analysis



IMPLMENTATION



AI Capstone

Logout

Welcome, Nexora Technologies

Welcome to your AI Capstone Company Dashboard.

Company Information

Company Name: Nexora Technologies

Email: Nexoratech1@gamil.com

Industry: Information Technology & Software

Location: Chennai, Tamil Nadu, India

Website: https://www.nexoratechnologies.com

Description:

NexaByte Technologies is an innovative technology company specializing in artificial intelligence, software development, cloud solutions, and digital transformation. We collaborate with students and professionals to develop practical technology solutions and provide opportunities for innovation and industry experience.

Post Capstone Project

Post a Capstone Project

Create a project opportunity for students to explore and apply.

Project Title

Example: AI-Based Healthcare Prediction System

Project Description

Describe the project, objectives and expected outcome...

Project Domain

Example: Artificial Intelligence / Healthcare

Technology / Skills Required

Example: Python, Machine Learning, React, MySQL

Duration

Example: 3 Months

Difficulty Level

Select Difficulty

Number of Students Required

Example: 5

Application Deadline

dd - mm - yyyy

Post Capstone Project

TESTING

- **Unit Testing** – Test individual modules such as login, dashboard and recommendation logic.
- **Integration Testing** – Check the connection between **Frontend** → **Spring Boot** → **MySQL**.
- **Functional Testing** – Verify buttons, forms, navigation and page functions.
- **Database Testing** – Check whether student, company, project and skill data are stored and retrieved correctly.
- **UI Testing** – Check page layout, responsiveness and user-friendly navigation.
- **AI Testing** – Test whether project recommendations and skill-gap results match the student's skills and interests.
- **Error Testing** – Verify proper handling of invalid login, missing data and incorrect inputs.

CHALLENGES

- ✓ **AI Recommendation Logic** – Developing accurate project matching based on skills and interests.
- ✓ **Skill Gap Detection** – Identifying missing skills required for each project.
- ✓ **Database Design** – Managing relationships between students, companies, projects and skills.
- ✓ **Two-Module Integration** – Connecting the Student and Company modules smoothly.
- ✓ **Testing & Debugging** – Resolving navigation, routing and integration errors during development.
- ✓ **Personalization** – Generating recommendations based on individual student profiles.

THANK
YOU